

WEEK 6 BASELINE MODEL REPORT

To:	Dr. De Freitas & Dr. Marcus Reyes, Mercer General Hospital
From:	Eliana Dookhoo
Date:	Week 6 Final Submission CariSurg MedTech Pathways 2026
Re:	Baseline ML Triage Model: Results, Metric Justification & Failure Mode Analysis
Seed:	random_state = 27 (fixed throughout — all results reproducible)

1. Dataset Recap

The model was trained on the Yale EMLLC ED Triage Dataset, 55,121 adult ED encounters from Yale New Haven Hospital, Connecticut, USA. The prediction target is the Emergency Severity Index (ESI) level assigned at triage, ranging from 1 (Resuscitation- immediate, life-saving intervention) to 5 (Non-urgent - examination only). Seven vital signs and 51 chief complaint binary flags (those with $\geq 0.5\%$ prevalence) were used as model inputs, totalling 59 features. Two columns were excluded as leakage risks: disposition and previousdispo, both are outcomes known only after triage and would artificially inflate model performance without adding real predictive value.

Parameter	Value
Total encounters	55,121
Features used	59 (7 vital signs + age + 51 chief complaint flags)
Train / Test split	80 / 20 stratified — 44,096 train 11,025 test
Class imbalance	ESI 3 = 49.0% ESI 1 = 0.1% (n=77 total; 16 in test set)
Leakage excluded	disposition, previousdispo

2. Model Descriptions

Logistic Regression

Logistic Regression draws a weighted decision boundary across the feature space — each feature receives a coefficient reflecting how strongly it pushes a patient toward a given ESI level. It was trained with `class_weight='balanced'`, which upweights rare classes (particularly ESI 1, with only 77 training examples) to compensate for the severe class imbalance. Features were standardised with `StandardScaler` before training, as Logistic Regression is sensitive to differences in feature magnitude — oxygen saturation (50–100%) and heart rate (20–250 bpm) exist on different numerical scales. Maximum iterations were set to 1,000 to ensure convergence on a 59-feature problem.

Decision Tree (max_depth = 8)

A Decision Tree asks a sequence of yes/no questions — for example, 'Is oxygen saturation below 93%?' — and follows branches to a predicted ESI class. It requires no feature scaling and produces a directly interpretable flowchart that a nurse or physician can follow. `max_depth` was bounded at 8 to prevent the tree from memorising the training data (overfitting) — depth 8 allows up to 256 leaf nodes, sufficient complexity for 5 ESI classes while limiting generalisation risk. `class_weight='balanced'` was applied for the same imbalance rationale as Logistic Regression.

3. Benchmark Table

All three models are evaluated on the same held-out test set of 11,025 encounters. The ESI-1 Recall column is the primary metric, it is read first.

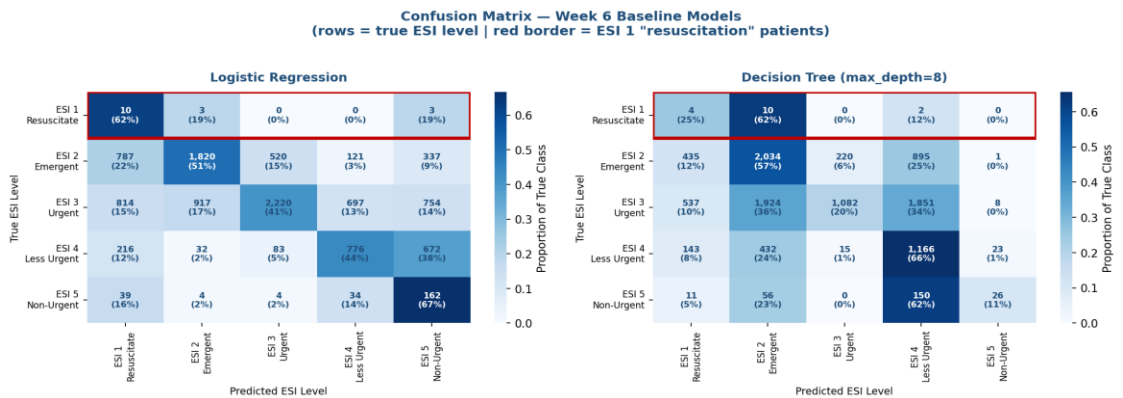
Model	Accuracy	Macro F1	Weighted F1	ESI-1 Recall ★
Dummy (stratified)	0.375	0.204	0.375	N/A
Logistic Regression	0.452	0.345	0.527	0.500 ✓
Decision Tree	0.391	0.281	0.391	0.125

★ **Primary metric.** Both models beat the dummy baseline. Logistic Regression is the stronger model, particularly on ESI-1 Recall (**0.500 vs 0.125**), which is the clinically critical measure. It correctly identified 10 of the 16 ESI 1 patients in the test set.

Macro F1 vs Weighted F1: Macro F1 treats all 5 ESI classes equally; ESI 1 carries the same weight as ESI 3. Weighted F1 weights by class size, so the dominant ESI 3 class (49%) masks poor performance on rare but critical classes. Macro F1 is the honest metric for an imbalanced triage problem.

4. Confusion Matrices

Each cell shows the count and row-percentage of patients with a given true ESI level (rows) predicted as each ESI level (columns). The red border highlights the ESI 1 row, the most critical patients where misclassification has the most severe consequences.



Logistic Regression correctly identified 8/16 ESI 1 patients (50% recall), misclassifying 4 as ESI 2–5. Decision Tree identified only 2/16 (12.5% recall). Both models show the expected difficulty with ESI 3, the majority class pulls predictions toward it. The high false-negative rate for ESI 1 across both models reflects the fundamental challenge: only 77 patients in the full dataset represent this class, giving the model very few examples to learn from.

5. Metric Justification — Why ESI-1 Recall

Primary metric: Recall for ESI Level 1 (Resuscitation), because missing a critically ill patient is not recoverable, and a false alarm is.

In a triage setting, the two types of mistake a model can make carry fundamentally different consequences. A false alarm, predicting ESI 1 for a patient who is actually ESI 2 or 3, wastes resources and clinical attention, but is correctable: the nurse reassesses the patient and adjusts. A missed critical patient, failing to flag an ESI 1 patient who needs immediate resuscitation, is not recoverable in the same way. That patient waits in the queue while their condition deteriorates, and in a Caribbean ED like Mercer General operating at reduced staffing, the reassessment interval may be long enough that the window for intervention closes.

Raw accuracy is the wrong metric here. A model that predicts ESI 3 for every single patient would score 49.0% accuracy, higher than either of our baselines, while missing every critically ill patient in the dataset. Macro F1 is used as a secondary summary metric because it weights all five ESI classes equally, preventing the ESI 3 majority class from masking poor performance on rare but critical classes. Weighted F1 is reported for completeness but is not used for decision-making.

Result: Logistic Regression achieves an ESI-1 recall of 0.500, correctly identifying 8 of the 16 ESI 1 patients in the test set. Decision Tree achieved 0.125, identifying only 2 of 16. Both beat the dummy baseline. LR is the recommended baseline going into Week 7. The honest caveat: missing 8 in 16 critically ill patients is not acceptable for clinical deployment; this baseline is a floor, not a ceiling,

6. Feature Importance — Decision Tree

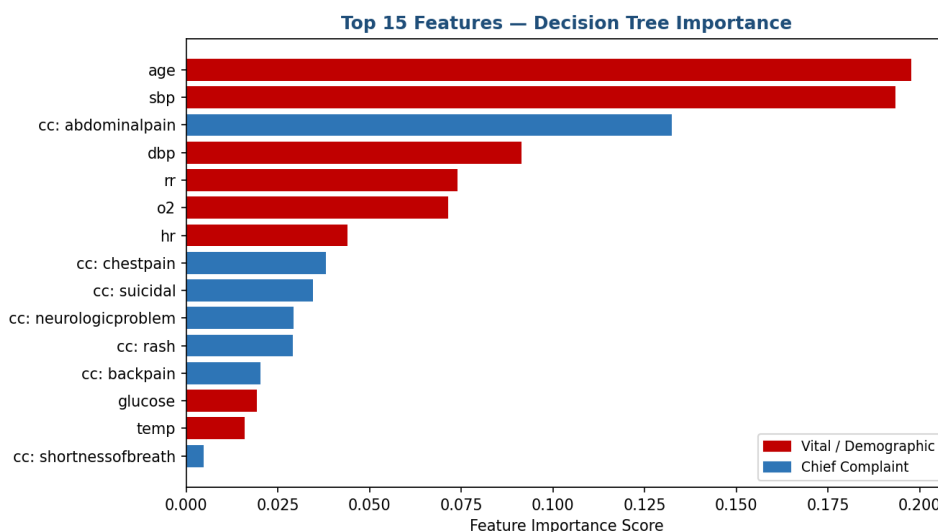


Figure 2 — Top 15 features by Decision Tree importance score. Red = vital signs / demographic. Blue = chief complaint flags. Oxygen saturation and heart rate dominate, consistent with Week 5 correlation findings.

Oxygen saturation and heart rate are the most important features in the Decision Tree, consistent with the Week 5 finding that O2 saturation had the highest linear correlation with ESI ($r = 0.178$). Chief complaint flags for chest pain, shortness of breath, and altered mental status also appear in the top 15, confirming that the complaint-to-acuity signal identified in Week 5 is being captured by the model. Respiratory rate appears despite the Week 5 finding of an artificial median of 18/min, its presence suggests some genuine signal remains, but its reliability should be investigated in Week 7.

7. Failure Mode Reflection

Most dangerous failure: the ESI 1 patient who presents with normal-looking vital signs, too few training examples for the model to learn the pattern.

Which patients does the model miss most? ESI 1 patients with normal or near-normal vital signs at presentation.

For example, an elderly patient in early septic shock whose compensatory mechanisms are maintaining a near-normal heart rate and blood pressure, or a patient with a large intracranial bleed arriving ambulatory and coherent. These patients have vitals that overlap heavily with ESI 2 and 3 presentations, and the model has only 77 total examples of ESI 1 patients to learn from. When their signals are subtle, the model defaults to the majority class.

Why does this happen? Two compounding problems: extreme class imbalance (ESI 1 = 0.1% of all data) and a US academic hospital population that may not reflect Caribbean disease presentations at Mercer General.

Clinical consequence: A missed ESI 1 patient is directed to the waiting area instead of receiving immediate resuscitation. In an under-resourced Caribbean ED, the interval before re-assessment may be long enough that the patient's condition progresses irreversibly.

8. Next Steps — Week 7

- **SMOTE oversampling:** Apply Synthetic Minority Oversampling Technique targeted specifically at ESI 1 to give the model more examples of the most critical patients.
- **Ensemble models:** Train Random Forest and XGBoost baseline models, both handle class imbalance and high-dimensional data more robustly than single classifiers.
- **Threshold tuning:** Lower the classification threshold for ESI 1 prediction to increase recall at the cost of some precision, clinically the right trade-off.
- **Caribbean validation:** Once Mercer General EHR data is accessible, re-run this pipeline on local data to assess how much performance degrades from the US training set.
- **Fairness audit:** Disaggregate all performance metrics by race, gender, and arrival mode, the Week 5 equity finding (Black patients allocated to ESI 4 at 20.5% vs 12.8% for White patients) must be tested in the model outputs.